Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior backend engineer with 6+ years shipping **production software** for distributed product teams in the US, Singapore and Uganda. Owns work end to end, from solution design to deployment: led a **zero-downtime migration of 20+ production applications** from Parse/MongoDB to Firebase/GCP, reporting to the CTO, with a live two-way sync between datastores while both carried traffic. Backend in **Node.js** (plus Python tooling), **integration with data services** and third-party APIs, containerised microservices on **Docker and Kubernetes**, and hands-on cloud operations on GCP and AWS. Led a team of 4 developers. Built a USSD service on carrier channels for legal-aid providers in Uganda. Now applying the same small-team, high-ownership approach to AI-backed telco products.

CORE COMPETENCIES

Backend Engineering (Node.js, Python) | End-to-End Solution Design | Integration with Data Services & APIs | Containers (Docker, Kubernetes) | Cloud Deployment (GCP, AWS) | Event-Driven Systems (Pub/Sub, Kafka, gRPC) | Zero-Downtime Migrations | Team Lead & Engineer Enablement

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote · e-commerce

- Owned the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Built a **real-time two-way data sync** between MongoDB and Firestore in Node.js on Google **Pub/Sub** message processing, keeping two live production systems consistent through the cutover.
- Migrated all services off **AWS** (S3, EC2, EBS), saving **$5,000/month**; migrated file storage from Amazon S3 to GCS with a **Python** script.
- Shipped product features end to end: Express Shipping (**+$40 revenue per order**) and a 3D visualisation feature (video overlay + ffmpeg) that increased buyer conversion; migrated a Webflow site from Parse to Firebase.

MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Built the migration script that read from MongoDB, ran complex processing logic, and wrote to Firestore.
- Wrote documentation for fellow engineers on the new Firebase SDKs (**Node.js and Python**); trained the Ops team on the dashboard and backup procedures.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote · fintech PWA

- Integrated third-party market data: built ad-hoc jobs over **2M+ Firestore records** to keep customer prices current from **Morningstar APIs**.
- Extended backend APIs in TypeScript on Firebase Cloud Functions and Express.js (Heroku); added responsive React/Tailwind pages.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote · agency

- Built backends for DeFi applications in Node.js with Web3; delivered React UIs from Figma designs under tight timelines, reporting to the CTO.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote · legal tech for NGOs

- Led a team of **4 developers**; architected a **microservices backend** for FIDA Uganda's case management app on **Apache Kafka, Docker and Kubernetes**.
- Built a **USSD service** for legal-aid providers communicating with a Node.js backend via **gRPC**; delivered a paralegal database for LASPNET and a React Native app with Firebase push notifications.

PROJECTS

Real-time Pub/Sub Sync Service

Two-way MongoDB↔Firestore sync running under parallel production traffic across 20+ applications during a zero-downtime migration.

Node.js · Google Pub/Sub · MongoDB · Firestore

Healthcare WhatsApp Assistant AI integration

Integrated model outputs into a product: a conversational assistant that reads prescription photos and responds to patients in WhatsApp, using OpenAI for dialogue and Google Vision for OCR, behind the WhatsApp Business API.

OpenAI API · Google Cloud Vision · WhatsApp Business API · Node.js

EDUCATION

B.Sc. Computer Engineering — Makerere University 2018 – 2022

Kampala, Uganda

SKILLS

Languages: JavaScript, TypeScript, Python, SQL, Go (intermediate), Rust (intermediate)

Backend: Node.js, Express.js, REST APIs, gRPC, Microservices, Firebase Cloud Functions

Data & Messaging: MongoDB, Firestore, Google Pub/Sub, Apache Kafka, NATS Streaming

Cloud / Infra: GCP, Firebase, AWS (S3, EC2, EBS — intermediate), Docker, Kubernetes, Git, Linux

AI Integration: OpenAI API, Google Cloud Vision **Frontend / Mobile:** React, React Native, Tailwind, HTML, CSS **Spoken:** English, Kiswahili